COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE010 / Data Structures and Algorithms | **SECTION:** | CPE21S3 |
| **DATE PERFORMED:** | 07/14/2026 | **DATE SUBMITTED:** | 07/14/2026 |
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## SEATWORK NO. 2.1 Memory Diagram Exercise

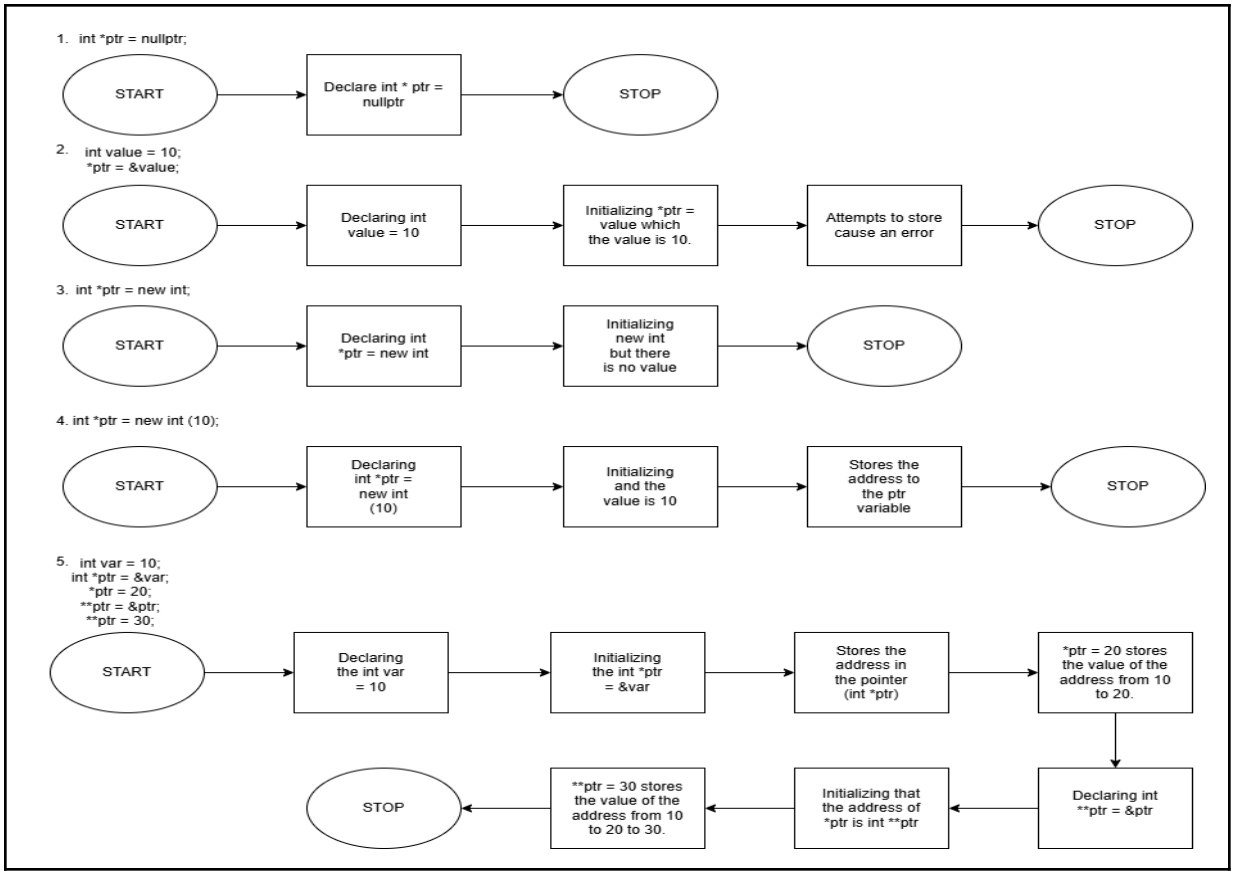
### 1. PROCEDURE OUTPUT

Make the diagram of the following:

```
1. int *ptr = nullptr;
2. int value = 10;
   int  *ptr= &value;
3. int *ptr = new int;
4. int *ptr = new int(10);
5. int var = 10;
   int *ptr = &var;
       *ptr = 20;
   int **ptr = &ptr;
       **ptr = 30;
```

### 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Diagram/Drawing/Flowchart:




## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **ChatGPT**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Idea generation**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **I used ChatGPT to give me info about pointers on a specific code especially #5 since it has the most codes out of all the questions given.**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.




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